

Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

Taihei Murakami, Kentaro Sasaki, Soichiro Oda, Kazuma Okada, and Yasuhiro Matsunaga

Supplementary methods

Additional details for the heterogeneous MD data set

The heterogeneous data set was assembled from a SAbDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. This rule yielded 1143 samples with a usable sequence, topology, trajectory, and native-contact label, representing 833 unique amino-acid sequences. Ninety-seven sequences occurred in more than one PDB entry and accounted for 407 samples. We calculated native-contact values separately for each structure and sampled them individually during training; duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Eleven sequences exceeded the 158-residue model input and were truncated at the end.

Structures were cleaned, missing atoms were added, and protonation was assigned at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [2, 3]. The termini were left uncapped. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4, 5]. The 300-K branch comprised 1 ns NVT and 1 ns NPT with a 2-fs timestep. Two 1-ns restrained NVT stages at 400 K preceded unrestrained 400-K NVT production scheduled for 100 ns. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps⁻¹ friction [6].

The contact calculation used all backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the first production frame, as for the mutation scans. Q was averaged over the final 30 ns of production, that is the last 300 frames of 100-ps output. The two MD data sets therefore used the same Q definition, contact set, native reference, and averaging window, and differed only in the variants they covered.

FEP variant and charge-change checks

The FEP tables used here comprised Ala, Asp, Gln, and Ile scans. Raw Phe scans existed but were not included in the processed tables, and the original reason for their omission was not recorded. Our conclusions therefore apply to the four-scan tables. For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [7], $0.086(\Delta q)^2$ kcal mol⁻¹. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the [0, 1] scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected $\Delta\Delta G$.

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the Tm-only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4} . The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5} . Both conditions used a non-encoder learning rate of 3×10^{-4} , weight decay 0.04, and ten training seeds.

References

- [1] Dunbar, J., Krawczyk, K., Leem, J., Baker, T., Fuchs, A., Georges, G., et al. SAbDab: the structural antibody database. *Nucleic Acids Research* 42, D1140–D1146 (2014). <https://doi.org/10.1093/nar/gkt1043>
- [2] Dolinsky, T. J., Nielsen, J. E., McCammon, J. A., Baker, N. A. PDB2PQR: an automated pipeline for the setup of Poisson–Boltzmann electrostatics calculations. *Nucleic Acids Research* 32, W665–W667 (2004). <https://doi.org/10.1093/nar/gkh381>
- [3] Olsson, M. H. M., Søndergaard, C. R., Rostkowski, M., Jensen, J. H. PROPKA3: consistent treatment of internal and surface residues in empirical pK_a predictions. *Journal of Chemical Theory and Computation* 7, 525–537 (2011). <https://doi.org/10.1021/ct100578z>
- [4] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguette, L., Huang, H., Migués, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. *Journal of Chemical Theory and Computation* 16, 528–552 (2020). <https://doi.org/10.1021/acs.jctc.9b00591>
- [5] Izadi, S., Anandakrishnan, R., Onufriev, A. V. Building water models: a different approach. *The Journal of Physical Chemistry C* 115, e000000_1

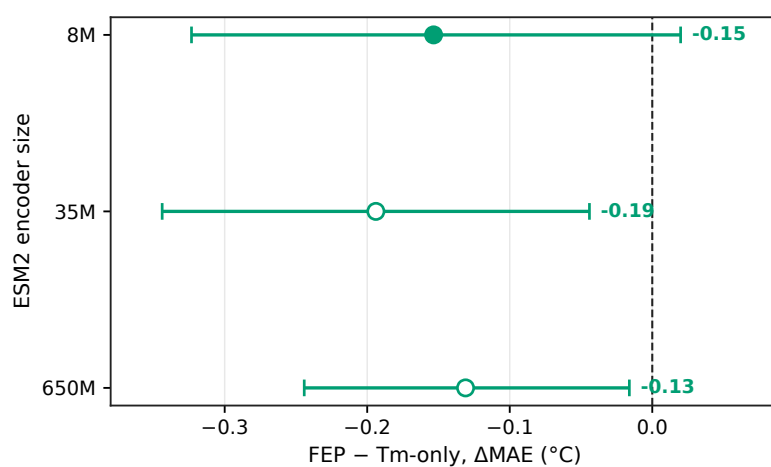


Figure S1 The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE (ΔMAE) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.

Chemistry Letters 5, 3863–3871 (2014). <https://doi.org/10.1021/jz501780a>

- [6] Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. PLOS Computational Biology 13, e1005659 (2017). <https://doi.org/10.1371/journal.pcbi.1005659>
- [7] Rocklin, G. J., Mobley, D. L., Dill, K. A., Hünenberger, P. H. Calculating the binding free energies of charged species based on explicit-solvent simulations employing lattice-sum methods: An accurate correction scheme for electrostatic finite-size effects. The Journal of Chemical Physics 139, 184103 (2013). <https://doi.org/10.1063/1.4826261>